Sequences

A *sequence* is a pattern of numbers or shapes.

1, +5→ 6, +5→ 11, +5→ 16, +5→ 21

Arithmetic Series

pattern : + OR -

2, ×3→ 6, ×3→ 18, ×3→ 54, ×3→ 162

Geometric Series

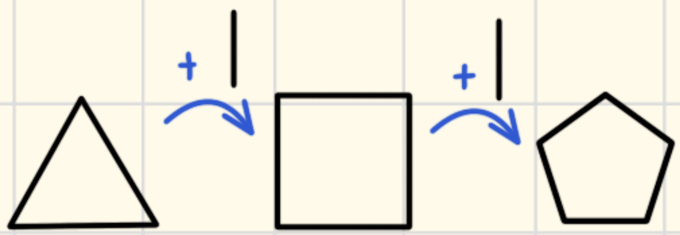
pattern : x OR ÷

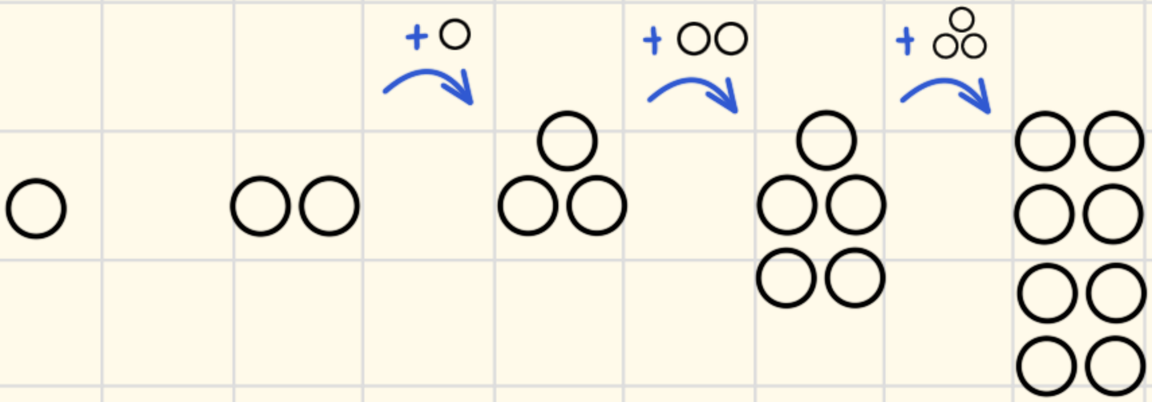
Each number is called a *term*.

Sequences

More sequences ...

4, +1 5, +2 7, +3 10





Sequences

A term in a sequence can be represented by a variable x_n with n the position of the term.

1, 6, 11, 16, 21

	1st term	2nd term	3rd term	4th term	5th term
n	1	2	3	4	5
x	x_1	x_2	x_3	x_4	x_5

Arithmetic Sequences

You can find the nth term of an arithmetic sequence by

$$a + (n - 1)d$$

- a: first term
- n: position of term
- d: difference of consecutive terms

Arithmetic Sequences

1) What is the 100th term of :

$$1, \quad 6, \quad 11, \quad 16, \quad 21$$

Solution

$a = 1$ first term

$d = +5$ difference

$n = 100$ position

$$a + (n-1)d$$

$$= 1 + (100-1)(5)$$

$$= 496$$

Arithmetic Sequences

2) An arithmetic sequence has

4th term = 25 and 7th term = 49

Find a and d.

Solution

The nth term : $a + (n-1)d$

$$a + 6d = 49 \quad n = 7$$
$$\ominus$$
$$a + 3d = 25 \quad n = 4$$

$$\rightarrow \quad 3d = 24$$

$$d = 8$$

$$a = 25 - 3d = 1$$

Geometric Sequences

You can find the nth term of an geometric sequence by

$$a \times r^{n-1}$$

- a: first term
- r: ratio of consecutive terms
- n: position of term

GEOMETRIC SEQUENCES

1) What is the 10th term of :

2, 10, 50, 250

Solution

$a = 2$ first term

$r = 5$ ratio

$n = 10$ position

$$a\,r^{n-1}$$

$$= 2 \times 5^{10-1}$$

$$= 3{,}906{,}250$$

Arithmetic Series

An arithmetic series is a sequence of the sums of an arithmetic sequence

$$S_n = \frac{n}{2}\left[2a + (n-1)d\right]$$

- S_n: Sum of the first n terms
- a: first term
- d: difference

Arithmetic Series

1) What is the sum of the first 100 terms of :

1 , 6 , 11 , 16 , 21

Solution

$a = 1$ first term

$d = +5$ difference

$n = 100$ position

$$S_n = \frac{n}{2}\left[2a + (n-1)d\right]$$

$$S_{100} = \frac{100}{2}\left[2(1) + (100-1)5\right]$$

$$= 50\left[497\right]$$

$$= 24,850$$